

WORK EXPERIENCE

AI/ML Engineer, KnowVia Tech Inc.

Mar 2026 – Present

Engineered Demand Forecasting Engine with MLflow and Scikit-Learn to optimize enterprise purchasing decisions

- Drove predictive AI strategies to resolve critical inventory inefficiencies, overcoming severe seasonal data anomalies and formatting inconsistencies that historically skewed enterprise demand forecasting.
- Built Python, Pandas and NumPy data pipelines containerized with Docker, processing 2M+ records, leveraging IQR outlier capping and 15 engineered features to train highly accurate Scikit-Learn (Random Forest/Linear Regression) algorithms across 50 retail categories.
- Reduced dead stock by 10,000+ units and expanded margins by delivering robust, mathematical forecasting insights to senior retail stakeholders, with model experiments tracked and versioned via MLflow.

Built End-to-End ML Architecture using MLflow and PyTorch to demystify tensor math and guide 3 cohorts

- Drove educational computer vision initiative using Python, Pandas, NumPy and MLOps to transform theoretical AI concepts into practical applications, mitigating critical dataset biases that degraded early model viability.
- Built scalable prediction pipelines using Docker and MLflow, preprocessing 10,000+ images and injecting 2,000 synthetic data points via image augmentation, directing end-to-end technical execution for 3 intern cohorts.
- Scaled model precision to 92% against unseen data batches using scikit-learn k=5 cross-validation, launching functional FastAPI proof-of-concept that eliminated beginner adoption barriers and proved technical feasibility.

Software Developer, Progress Solutions

Aug 2025 – Mar 2026

Designed FinTech ML and Data Model utilizing SQL, Scikit-Learn, and dbt to optimize loan approval rates

- Engineered predictive loan evaluation model using Python, Scikit-Learn, and XGBoost to prevent millions in banking losses, identifying high-risk applicants across 21 complex financial attributes.
- Designed reproducible data pipeline utilizing PostgreSQL and dbt to structure star-schema data warehouses, optimized model performance by implementing a custom cost matrix that penalized false positives 5x.
- Deployed model into production using Docker and AWS, establishing scalable architecture for real-time inference and cross-functional operations, aligning AI performance with core business profitability metrics.

Orchestrated Telemetry Aggregation Platform using Docker, Airflow, SQL to eliminate data silos and reduce audit hours

- Architected an automated AWS data pipeline, containerizing REST API integrations via Docker and Amazon ECS to securely extract and unify data from 4 isolated enterprise sub-systems, eliminating data silos.
- Engineered Python, Pandas, and SQL ETL workflows, orchestrating nightly DAGs via Amazon MWAA and Apache Airflow to stage S3 data and enforce 100% ISO 8601 date-time compliance.
- Saved 40+ hours in monthly manual audits and transformed asset visibility by centralizing an enterprise portfolio of 25,000+ infrastructure records into a scalable AWS-to-Snowflake architecture.

Software Engineer, Tata Consultancy Services

Oct 2021 – Jul 2023

Implemented Fault-Tolerant AI Routing Engine using TF-IDF and FastAPI to slash user wait times

- Drove technical resolution for critical dispatching delays impacting IT operations overwhelmed by highly variable, unstandardized user support requests.
- Architected a Python-based NLP text normalization pipeline using spaCy, SVM, and TF-IDF, deploying robust regex logic to minimize jargon-induced model misclassifications across a 50,000+ record dataset.
- Automated routing of 500+ daily tickets via REST API integrations built with FastAPI and deployed via Docker, accelerating time-to-resolution for end-users and permanently eliminating \$40,000 in annual operational costs.

Engineered Data Harmonization Pipeline using dbt, Snowflake to scrub records and reduce downstream supply errors

- Owned E2E resolution of critical SAP S/4HANA ERP data discrepancies across Nestlé's supply chain operations, designing Tableau dashboards to expose data quality gaps and align users and global stakeholders.
- Architected automated data pipelines using PostgreSQL, dbt, and Apache Airflow to scrub 1.5M+ legacy records and deployed complex fuzzy-matching routing logic leveraging Python and Pandas for 300+ raw materials.
- Reduced data-related supply errors by 33% by overhauling core Snowflake data models, providing clean, real-time data feeds that eliminated severe factory forecasting failures and prevented vital stock-outs.

Other Professional Experience

Software Developer, *User Genome – CodeXile*

Sep 2020 – Aug 2021

Engineered highly available AWS EC2 and Flask backend with a React frontend, leveraging MongoDB 2dsphere indexing and OAuth 2.0 to slash driver-matching geospatial latency from 45s to 5s and securely scale to 5,000+ active sessions.

PROJECT WORK

Engineered AI Anti-Spoofing Pipeline to optimize video ingestion and trigger security alerts

Aug 2024 – Dec 2024

- Engineered live Dockerized authentication system to combat bypass attempts, optimizing video ingestion pipeline with OpenCV buffer management, NumPy and Python memory profiling to resolve hardware-crashing leaks.
- Deployed 3D optical flow model using TensorFlow and Dlib to track 68 facial landmarks, achieving <0.1% False-Acceptance Rate across 5,000+ validation frames at 30 FPS, maintaining 100% uptime, and triggering asynchronous SMTP alerts within 2 seconds.

EDUCATION

University of Alabama at Birmingham (UAB) | *M.S. Computer Science* | GPA - 3.8/4

Kommuri Pratap Reddy Institute of Technology (KPRIT) | *B. Tech. Computer Science* | CGPA - 8.13/10

SKILLS, CERTIFICATIONS & LEADERSHIP

Skills: Python, C/C++, JavaScript, PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, OpenCV, spaCy, Dlib, MLflow, CNNs, AWS (EC2, S3, Lambda, SageMaker, ECS, CloudFront, CDK), Docker, Apache Airflow, Snowflake, dbt, PostgreSQL, MySQL, MongoDB, Deep Learning, Redis, FastAPI, React, REST APIs, Microservices Architecture, CI/CD, Git, GitHub, Jira, Linux, System Design, Biometric Tracking, Latency Optimization, Multi-threading, Memory Management

Certifications: Advanced Machine Learning, Deep Learning Specialization, AWS Cloud Practitioner, Python Specialization

Leadership: “Mastering MLOps: A Comprehensive Guide” | Author | [Medium](#)

Mar 2025

Academic Student Leader | UAB

Dec 2023 – Jan 2025

“New Method for Multi-hop Communication Over Bluetooth in Smartphones” | Co-Author | [IJECS](#)

Nov 2018